



NYC AIRBNB

Project: AI Programming Foundations



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1. Overview

This project develops a complete and reproducible data-analysis workflow using the NYC Airbnb Open Data dataset from 2019. The workflow includes data ingestion, data-quality assessment, cleaning, exploratory data analysis, visualization, interpretation, and responsible data practices. The project was designed as a reusable foundation for future machine learning, deep learning, generative AI, and agentic AI projects. The workflow emphasizes reproducibility through modular Python source files, a Jupyter Notebook, a documented Python environment, and Git version control, consistent with principles discussed by Danchev (2022).

2. Dataset Description

The dataset used for this project is the **NYC Airbnb Open Data** dataset from 2019. It contains **48,895 Airbnb listings and 16 variables** describing listing characteristics, hosts, locations, room types, prices, reviews, and availability. The five NYC boroughs represented are Manhattan, Brooklyn, Queens, the Bronx, and Staten Island. Key variables examined in this project include `neighbourhood_group`, `neighbourhood`, `room_type`, `price`, `number_of_reviews`, `last_review`, `reviews_per_month`, `minimum_nights`, and `availability_365`. The analysis focuses primarily on listing price, room type, borough, and review-related data quality.

3. Workflow Description

The project follows a sequential data workflow beginning with ingestion and ending with interpretation and documentation.

Ingestion

The raw CSV file, `AB_NYC_2019.csv`, is loaded into a Pandas DataFrame using `pd.read_csv()`. The raw dataset is retained as the source dataset and is not overwritten during cleaning. Initial inspection includes displaying the first rows, checking dataset dimensions, examining column names, and inspecting data types.

The dataset was confirmed to contain **48,895 rows and 16 columns**.

Cleaning

Before modifying the data, missing values and duplicate rows were examined. The analysis identified missing values in `last_review`, `reviews_per_month`, `name`, and `host_name`. No duplicate rows were found.

Two reusable cleaning functions were implemented in `src/data_cleaning.py`. The first function, `clean_review_fields()`, converts `last_review` to a datetime format and replaces missing `reviews_per_month` values with zero when a listing has no reviews. The second function, `clean_text_fields()`, replaces missing listing and host names with descriptive placeholder values.

The cleaning functions preserve all **48,895 observations** rather than deleting listings with missing values.

Exploratory Analysis

The exploratory analysis is implemented as a reusable function in `src/eda.py`. The `summarize_listings()` function produces dataset dimensions, descriptive price statistics, room-type counts, borough counts, average price by room type, and average price by borough.

Visualizations

Three visualizations were created using Seaborn and Matplotlib:

1. **Figure 1:** Distribution of Airbnb Listing Prices
2. **Figure 2:** Average Airbnb Price by Borough
3. **Figure 3:** Airbnb Prices by Room Type

The visualizations were selected to communicate the distribution of prices and compare prices across important categorical variables. Cleveland and McGill (1984) demonstrated the importance of graphical perception in how accurately people extract quantitative information from visual displays. The use of straightforward histograms, bar charts, and boxplots in this project therefore supports relatively direct comparisons between distributions and categories.

Summary

The final stage of the workflow interprets the findings, discusses limitations and potential sources of bias, and documents how another user could reproduce the analysis.

4. Key Decisions and Assumptions

Cleaning Decisions

The most important cleaning decision involved the missing review-related fields. The dataset contained **10,052 listings with zero reviews**, and all 10,052 of those listings had missing values for both `last_review` and `reviews_per_month`. At the same time, no listings with existing reviews were missing either of those fields.

Because the missingness was directly associated with having zero reviews, missing `reviews_per_month` values for these listings were replaced with zero. This represents the absence of monthly review activity rather than an unknown numerical measurement.

The missing `last_review` values were intentionally preserved. A listing with no reviews does not have a meaningful review date, so assigning an artificial date would introduce information that was not present in the original dataset.

Missing values in `name` and `host_name` were handled differently. They were replaced with "Unknown Listing" and "Unknown Host" rather than removing those observations. This preserved the affected listings and avoided unnecessary loss of data.

EDA Focus

The exploratory analysis focused on price, borough, and room type because these variables provide clear descriptive information about the Airbnb marketplace represented in the dataset. Price statistics were used to understand the central tendency and spread of listing prices. Grouped summaries were then used to compare average prices across room types and boroughs.

The mean price was interpreted alongside the median because the dataset contains substantial price variation. The mean listing price was **\$152.72**, while the median was **\$106**, indicating that high-priced listings pull the mean upward.

Visualization Decisions

Figure 1 uses a histogram with a KDE curve to show the overall price distribution. A **\$500 cutoff** was applied only to the visualization because extreme values up to \$10,000 can compress the majority of the observations and make the main distribution difficult to examine.

Figure 2 uses a bar chart to compare average listing prices across boroughs. This provides a direct comparison of the group means.

Figure 3 uses a boxplot to compare price distributions by room type. The boxplot makes it possible to examine differences in typical prices and variation within each room category.

The \$500 cutoff used in Figures 1 and 3 does not modify the cleaned dataset. Listings above \$500 remain in the underlying analysis data.

5. Results and Interpretation

Overall Price Distribution

The dataset contains **48,895 listings**. The mean listing price is **\$152.72**, while the median is **\$106**. The standard deviation is **\$240.15**, with prices ranging from **\$0 to \$10,000**.

The difference between the mean and median indicates that listing prices are strongly right-skewed. Most listings are concentrated toward the lower end of the price range, while a smaller number of expensive listings increase the mean.

Figure 1 illustrates this pattern by limiting the displayed range to listings priced at \$500 or below. This visualization cutoff allows the main distribution to be examined without removing or changing the underlying observations.

Room Type

The largest room-type category is **Entire home/apt**, with **25,409 listings**. Private rooms account for **22,326 listings**, while shared rooms account for **1,160 listings**.

Average prices differ considerably across these categories. Entire homes or apartments have an average price of approximately **\$211.79 per night**. Private rooms have an average price of approximately **\$89.78**, while shared rooms have an average price of approximately **\$70.13**.

Figure 3 shows these differences while also demonstrating that there is substantial variation within each room type. This means room type is associated with price differences, but room type alone does not explain all price variation.

Borough

Manhattan contains the largest number of listings, with **21,661**, followed by Brooklyn with **20,104**. Queens contains **5,666 listings**, the Bronx contains **1,091**, and Staten Island contains **373**.

Average prices also vary by borough. Manhattan has the highest average listing price at approximately **\$196.88 per night**. Brooklyn follows at approximately **\$124.38**, while Staten Island averages approximately **\$114.81**. Queens averages approximately **\$99.52**, and the Bronx has the lowest average at approximately **\$87.50**.

Figure 2 visually demonstrates these geographic differences. The results suggest that borough location is associated with differences in average Airbnb listing prices within this dataset. However, the analysis does not establish that location itself causes the observed price differences.

Review-Related Missing Data

The data-quality analysis identified a systematic relationship between missing review information and zero reviews.

There were **10,052 listings with zero reviews**, and all 10,052 had missing `last_review` and `reviews_per_month` values. There were no listings with reviews that were missing either of these fields.

This pattern supports treating missing `reviews_per_month` as zero for listings with no reviews while retaining missing `last_review` values.

6. Responsible Practice: Bias and Data Quality

Data cleaning decisions can influence analytical conclusions. Removing every row containing a missing value could have reduced the dataset substantially and potentially introduced selection bias. Instead, this project attempted to distinguish between missing values that represented meaningful conditions and missing values that could be safely replaced.

The review-related missing values provide an example of why missing data should be investigated before being automatically removed or imputed. Because all listings with zero reviews had missing review-frequency values, replacing those specific values with zero reflects the observed structure of the dataset.

However, the decision should not be generalized automatically to other datasets. Missing values may have different meanings depending on how the data was collected.

The replacement of missing listing and host names with descriptive placeholders preserves observations but does not recover the missing information. These placeholders should therefore not be interpreted as actual names.

The dataset also has potential representation bias. It describes Airbnb listings rather than the entire housing, hotel, or lodging market in New York City. Properties that were not listed on Airbnb are not represented. Consequently, the results should be interpreted as descriptive patterns among the listings included in the dataset rather than as estimates of the entire New York City housing market.

The dataset is also from **2019**, so the results should not be interpreted as a representation of current Airbnb prices or availability. Market conditions, regulations, demand, supply, and platform behavior may have changed since the dataset was collected.

Another limitation concerns extreme prices. The dataset includes listings with a price of \$0 and listings priced as high as \$10,000 per night. These observations may represent unusual cases, data-entry issues, or legitimate extreme listings. This project did not remove them from the underlying dataset. Instead, the \$500 cutoff was used only for selected visualizations.

Future analysis should investigate these unusual observations and compare conclusions using additional robust measures such as medians or alternative methods for handling extreme values.

7. Reproducibility

Reproducibility was an important design objective of this project. Danchev (2022) presents reproducible data science as a workflow involving open tools, computational practices, and organized analysis that allows data-science work to be rerun and examined by others. Following this principle, the project separates raw data, reusable Python functions, notebook analysis, visualization outputs, and environment dependencies.

The raw dataset is stored in:

`data/AB_NYC_2019.csv`

Reusable processing functions are stored in:

`src/data_cleaning.py`

The exploratory-analysis function is stored in:

`src/eda.py`

The visualization functions are stored in:

`src/visualizations.py`

The complete analysis notebook is stored in:

`notebooks/data_workflow.ipynb`

A rendered HTML version of the notebook is also included as:

`notebooks/data_workflow.html`

The visualization outputs are stored in:

outputs/figures/

The Python environment dependencies are documented in:

requirements.txt

Another user can install the documented dependencies with:

```
pip install -r requirements.txt
```

The notebook can then be opened with Jupyter Notebook or JupyterLab and executed from beginning to end.

Git is also used for version control. Multiple commits document the development of the project, while an additional development branch demonstrates organized version control and provides a separate environment for development work.

This separation of raw data, reusable code, analysis, outputs, and dependencies makes the project easier to reproduce and maintain.

8. Conclusion

This project demonstrates a complete data workflow using the 2019 NYC Airbnb Open Data dataset. The workflow moves from raw-data ingestion through quality assessment, cleaning, exploratory analysis, visualization, interpretation, and responsible data practices.

The analysis found substantial differences in average listing prices by both room type and borough. Entire homes or apartments had the highest average price among the room types examined, while Manhattan had the highest average listing price among the five boroughs. The overall price distribution was strongly right-skewed, demonstrating why the median and distribution should be considered alongside the mean.

The project also demonstrated the importance of examining missing data before applying cleaning rules. The systematic relationship between zero reviews and missing review fields supported a targeted cleaning approach that preserved all observations.

Finally, the project establishes a reusable structure for future AI and data-science work. Modular source code, a documented environment, reproducible notebook execution, visualization outputs, and Git version control provide a foundation that can later be extended to predictive modeling, deep learning, generative AI, and agentic workflows.

References

Cleveland, W. S., & McGill, R. (1984). Graphical perception: Theory, experimentation, and application to the development of graphical methods. *Journal of the American Statistical Association*, 79(387), 531–554. <https://doi.org/10.1080/01621459.1984.10478080>

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>

GitHub Repository

The complete project, including the Jupyter Notebook, reusable Python modules, dataset, visualization outputs, README, and reproducibility files, is available at:

[pravinganesh/ai-programming-foundations-project: AI programming foundations Project](#)